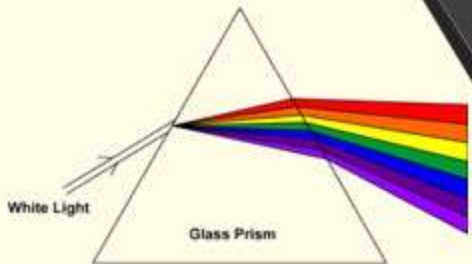


PHYSICS

SPHERICAL LENS

MODULE - 7



SIMPLE TRICK - HAVE YOU EVER TRIED THIS?



- **What happens** - A piece of paper catches fire when a magnifying glass is held over it.
- **Why it happens** - The magnifying glass actually converges the rays of sun into a tiny dot in that paper. So basically the point of the ignition is actually a very diminished but a real image of the Sun !



WHAT DO THESE PICTURES SUGGEST ?



Human eye



Reading glasses



Telescopic glasses



Camera lens

These all are made up with spherical lens.



HOW IS LENS DIFFERENT THAN A REGULAR GLASS ?

GLASS SLAB

- A glass slab can only laterally deviate light
- All the rays that pass through the lens deviate laterally but they never meet at a point.



LENS

- A lens is different than glass because it can focus light.
- All the rays that pass through the lens come out bent and they can meet at a point to form a sharp image.

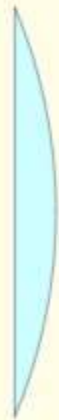


TYPES OF SPHERICAL LENSES

CONVERGING LENS



biconvex

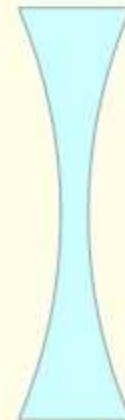


plano
convex



convex
meniscus

DIVERGING LENS



biconcave



plano
concave



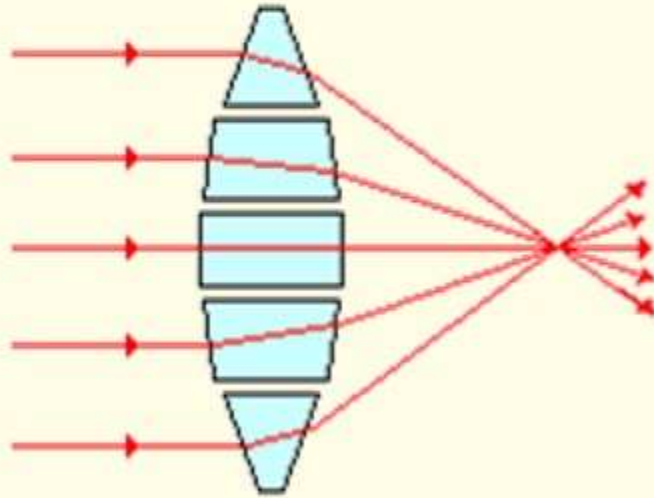
concave
meniscus



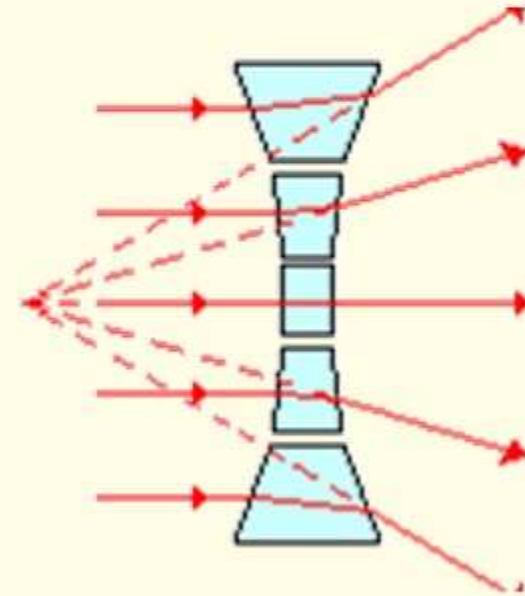


ANATOMY OF THE TWO DIFFERENT SPHERICAL LENSES

CONVERGING PRISMS



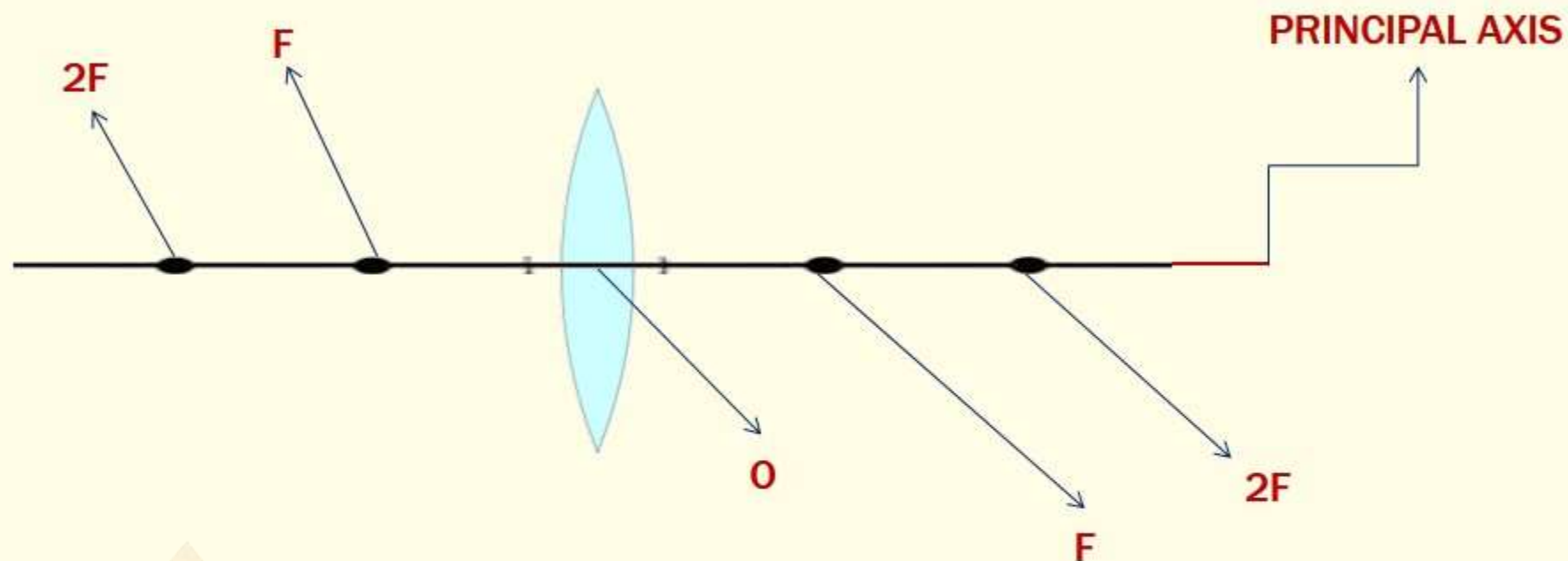
DIVERGING PRISMS



A set of prisms acting as a converging and diverging lens. When these prisms act together they produce a bright image focused at a point



SPHERICAL LENS - SOME IMPORTANT POINTS

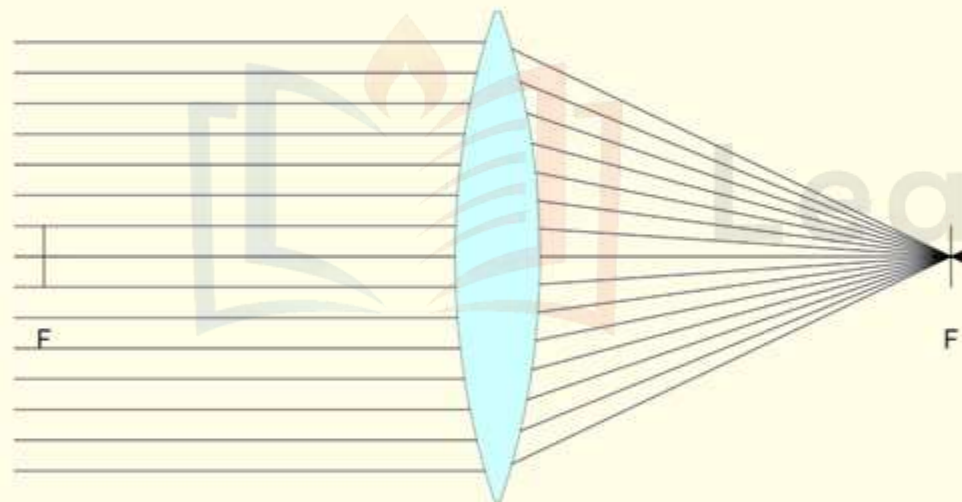


- F - focus
- $2F$ - Centre of curvature. The center of the big spherical lens
- O - Optical center
- O to F - Focal length





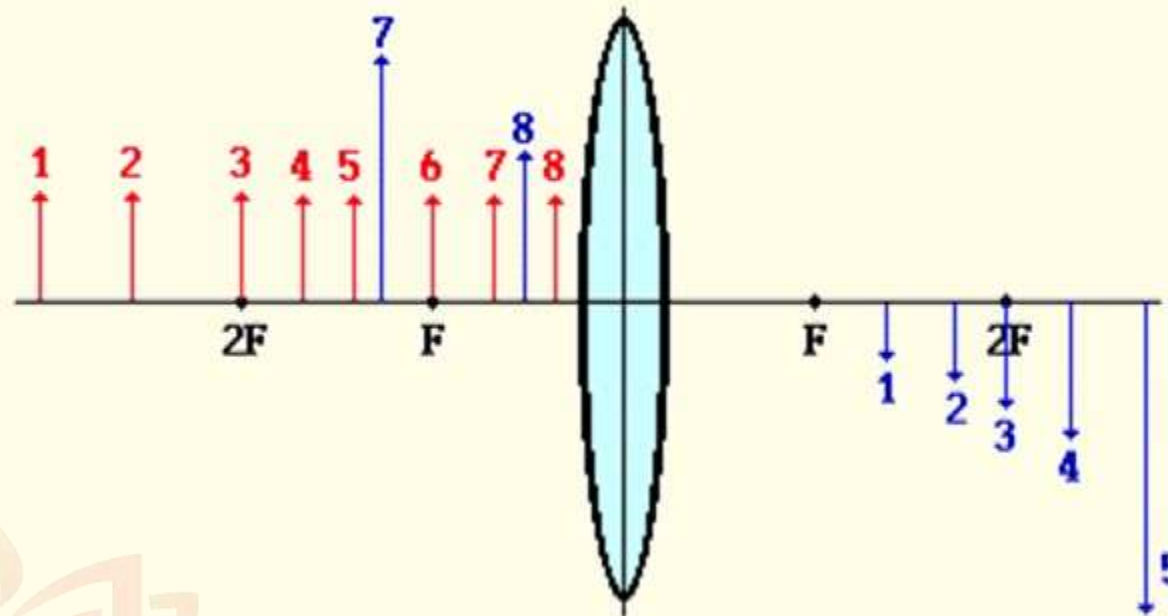
CONVEX LENS



- Thick at the center and thinner at the edges.
- It converges a parallel beam of light, which after refraction, comes and meets at point F (focus)
- A convex lens forms a real, inverted and diminished image of an object which is situated far away.
- The image starts to magnify when the object moves closer to the lens from the center of curvature.



OBJECT – IMAGE RELATIONS IN A CONVEX LENS

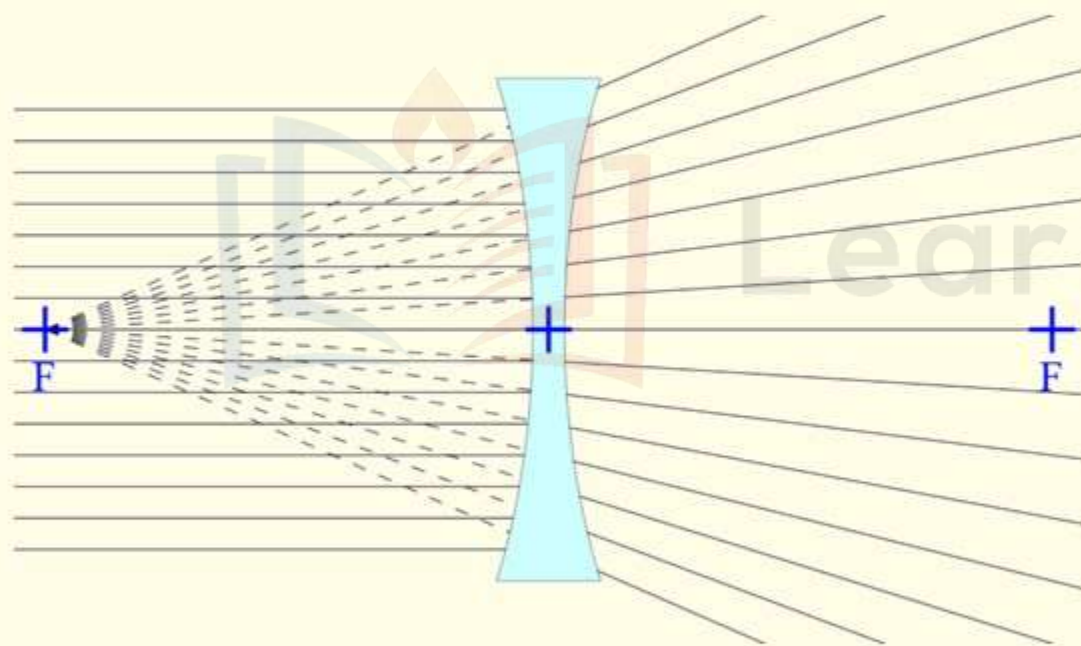


Only #6 does not have any image because the rays join at infinity and the image cannot be obtained





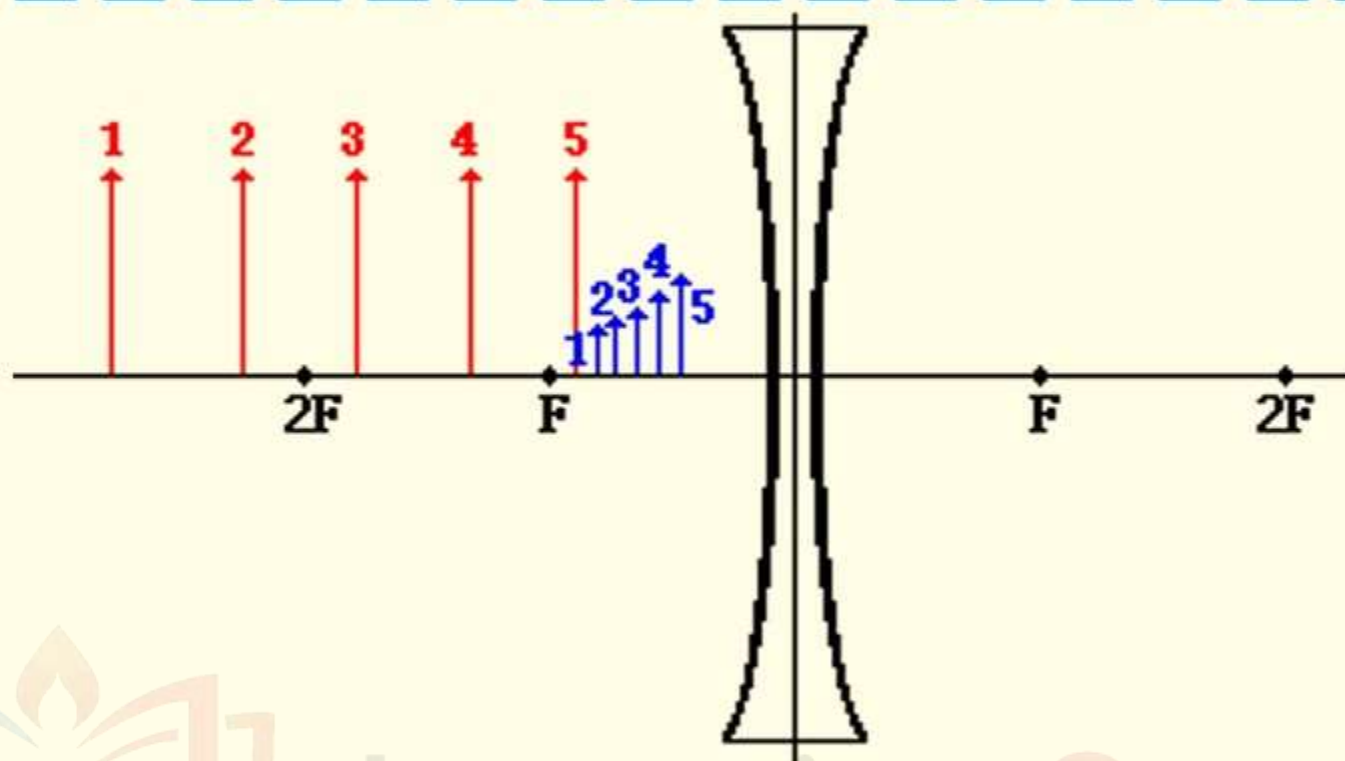
CONCAVE LENS



- Concave lens is a diverging lens
- Concave lens diverges rays of light that are traveling parallel to its principal axis.
- Diverging lenses are relatively thin across their middle and thick at their upper and lower edges.
- A concave lens always produces a virtual image because the rays seem to converge at a point while they are not.
- The image formed will be erect and diminished.



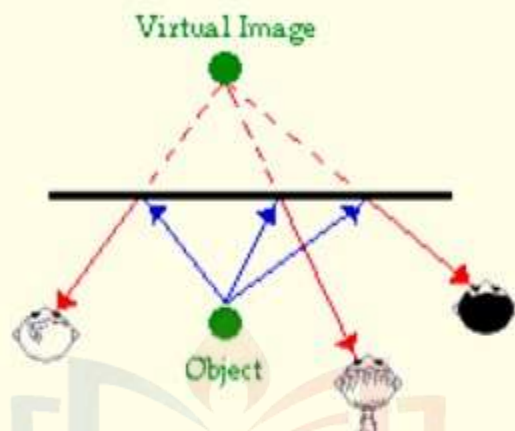
OBJECT – IMAGE RELATIONS IN A CONCAVE LENS



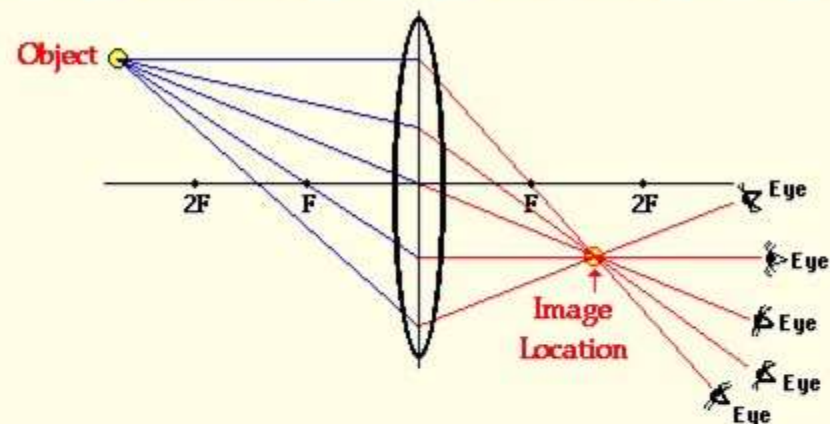
All images by a concave lens are formed virtual, erect and diminished in size.



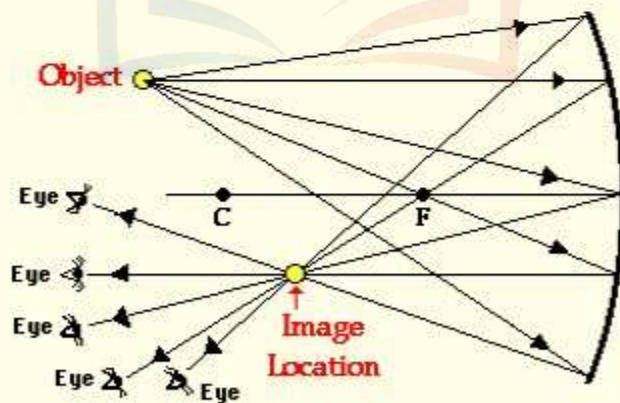
RECAP OF THE TYPES OF IMAGE FORMATIONS



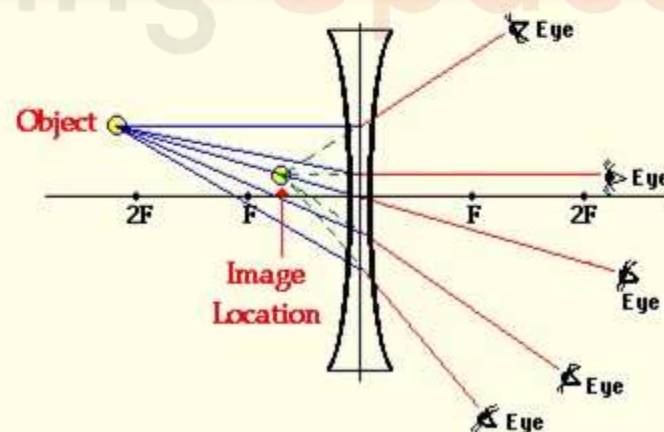
PLANE MIRROR



CONVEX LENS



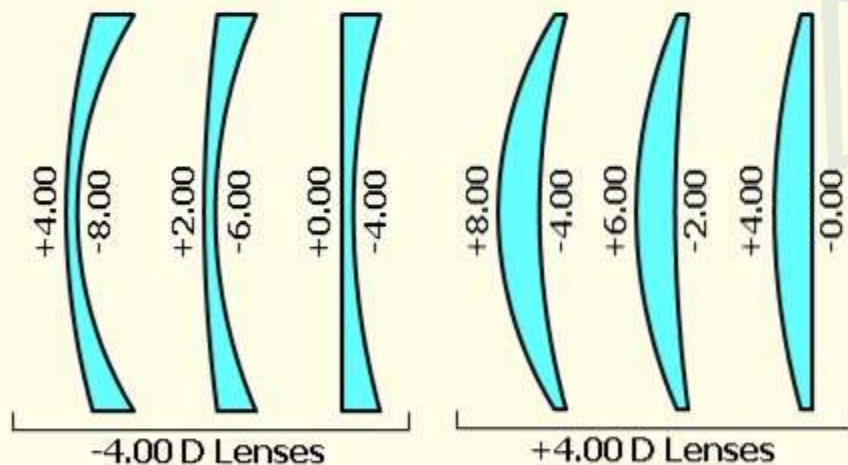
CURVED MIRROR



CONCAVE LENS



POWER OF A LENS



- Power of a lens is defined as the reciprocal of its focal length and the unit is expressed in Meters.
- Therefore, a lens of higher focal length has a low power and a lens of lower focal length has high power.
- If the curvature or bulge increases, the power increases.
- The SI Unit of Power is Diopetre (D)
- + type** is for convex or converging lens and **- type power** is for concave lens.





CONTRAST BETWEEN CONVEX AND CONCAVE LENS

CONVEX LENS

- Converging lens
- After refraction, light rays meet at one point or appear to meet.
- The image formed can be real, virtual, enlarged or diminished based upon the position of the object.
- Focal length is **positive**.
- Used to correct **Hypermetropia**

CONCAVE LENS

- Diverging lens
- After refraction, light rays move away from one another.
- The image formed is always virtual and diminished and the image cannot be obtained on a screen.
- Focal length is **negative**.
- Used to correct **Myopia**.



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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